Category Theory and Functional Programming
Appendix

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Preliminaries

Textbook

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Outline

Monoids

Definition

Let M be a set and let $(-) \cdot (-)$ be a binary relation on M and $1 \in M$. The structure $(M, \cdot, 1)$ is a **monoid** iff it satisfies

$$\forall x \forall y \forall z ((x \cdot y) \cdot z = x \cdot (y \cdot z)) \qquad \text{(associativity)}$$

$$\forall x (x \cdot 1 = x = 1 \cdot x) \qquad \text{(identity)}$$

Monoids

Example

The structure $(\mathbb{N}, +, 0)$ is a monoid.

Free Monoid

Definition

Let Σ be an alphabet (a set), let Σ^* be the set of strings over Σ including the empty string ε, and let $(-) \cdot (-)$ be the concatenation of strings. Then $(\Sigma^*, \cdot, \varepsilon)$ is the **free monoid** on the set Σ.

Monoid Homomorphisms

Definition

A **homomorphism** between monoids is a map between the domains of the monoids that preserves the monoid operation and the identity element.

Monoid Homomorphisms

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A **homomorphism** between monoids is a map between the domains of the monoids that preserves the monoid operation and the identity element.

Let $(M, \cdot, 1_M)$ and $(N, *, 1_N)$ be two monoids. A homomorphism from $(M, \cdot, 1_M)$ to $(N, *, 1_N)$ is a function $h : M \to N$ such that for all x, y in M:

$$h\,(x \cdot y) = h\,x * h\,y,$$
$$h\,(1_M) = 1_N.$$

Outline

Groups

Definition

Let G be a set, $(-) \cdot (-)$ be a binary relation on G and $1 \in G$. The structure $(G, \cdot, 1)$ is a **group** iff it satisfies

$$\forall x \forall y \forall z ((x \cdot y) \cdot z = x \cdot (y \cdot z)) \qquad \text{(associativity)}$$

$$\forall x (x \cdot 1 = x = 1 \cdot x) \qquad \text{(identity)}$$

$$\forall x \exists x' (x \cdot x' = 1 = x' \cdot x) \qquad \text{(inverse)}$$

Groups

Example

The structure $(\mathbb{Z}, +, 0)$ is a group.

Groups

Example

The structure $(\mathbb{Z}, +, 0)$ is a group.

Example

The monoid $(\Sigma^*, \cdot, \varepsilon)$ is not a group.

Direct Product

Definition

Let $(G, *, 1_G)$ and $(H, \diamond, 1_H)$ be two groups. The **direct product** of G and H is the group $(G \times H, \cdot, (1_G, 1_H))$ where

$$\begin{aligned}(-) \cdot (-) &: G \times H \to G \times H \\ (g_1, h_1) \cdot (g_2, h_2) &:= (g_1 * g_2, h_1 \diamond h_2).\end{aligned}$$

Direct Product

Definition

Let $(G, *, 1_G)$ and $(H, \diamond, 1_H)$ be two groups. The **direct product** of G and H is the group $(G \times H, \cdot, (1_G, 1_H))$ where

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Exercise

Show that the direct product of two groups is a group.

Outline

Algebraic Structures

Definition

An **algebraic structure** on a set $A \neq \emptyset$ is essentially a collection of n-ary operations on A (Birkhoff 1946, 1987).

Algebraic Structures

Description

A **homomorphism** is a structure-preserving map between two algebraic structures.

Algebraic Structures

Definition

A homomorphism φ between two algebraic structures is (Cohn 1981):

- a **monomorphism** if φ is an injection,
- an **epimorphism** if φ is a surjection,
- an **endomorphism** if φ is from an algebraic structure to itself,
- an **isomorphism** if φ is a bijection,
- an **automorphism** if φ is a bijective endomorphism.

Outline

Pre-orders

Definition

Let P be a set and let $\preccurlyeq$ be a binary relation on P. The relation $\preccurlyeq$ is a **pre-order** (or **quasi-order**) iff it satisfies

$$\forall x(x \preccurlyeq x) \qquad \text{(reflexivity)}$$

$$\forall x \forall y \forall z(x \preccurlyeq y \preccurlyeq z \Rightarrow x \preccurlyeq z) \qquad \text{(transitivity)}$$

The pair $(P, \preccurlyeq)$ is a **pre-ordered set** (or **quasi-ordered set**).

Pre-orders

Example

The pair $(\mathbb{N}, \leq)$ is a pre-ordered set.

Pre-orders

Example

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Example

The pair $(\{*\}, \{(*, *)\})$ is a pre-ordered set.

Pre-orders

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Question

Is the pair $(\emptyset, \emptyset)$ a pre-ordered set?

Pre-orders

Example

The pair $(\mathbb{N}, \leq)$ is a pre-ordered set.

Example

The pair $(\{*\}, \{(*, *)\})$ is a pre-ordered set.

Question

Is the pair $(\emptyset, \emptyset)$ a pre-ordered set?

Answer

Yes!

Pre-orders

Definition

Let $(S, \preccurlyeq_S)$ and $(T, \preccurlyeq_T)$ be two pre-ordered sets. A **homomorphism** from $(S, \preccurlyeq_S)$ to $(T, \preccurlyeq_T)$ is a function $h : S \to T$ such that, for all $x, y \in S$,

$$x \preccurlyeq_S y \quad \text{implies} \quad h\,x \preccurlyeq_T h\,y.$$

Pre-orders

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$$x \preccurlyeq_S y \quad \text{implies} \quad h\,x \preccurlyeq_T h\,y.$$

That is, a homomorphism from $(S, \preccurlyeq_S)$ to $(T, \preccurlyeq_T)$ is a monotone map $h : S \to T$.

Outline

Partial Orders

Definition

Let P be a set and let $\preccurlyeq$ be a binary relation on P. The relation $\preccurlyeq$ is a **partial order** iff it satisfies

$$\begin{aligned} &\forall x(x \preccurlyeq x) && \text{(reflexivity)} \\ &\forall x \forall y(x \preccurlyeq y \preccurlyeq x \to x = y) && \text{(anti-symmetry)} \\ &\forall x \forall y \forall z(x \preccurlyeq y \preccurlyeq z \to x \preccurlyeq z) && \text{(transitivity)} \end{aligned}$$

The pair $(P, \preccurlyeq)$ is a **partially ordered set** (or **poset**).

Partial Orders

Example

The pre-ordered sets $(\mathbb{N}, \leq)$, $(\emptyset, \emptyset)$ and $(\{*\}, \{(*, *)\})$ are posets.

Partial Orders

Question

Are pre-ordered sets which are not posets?

Partial Orders

Question

Are pre-ordered sets which are not posets?

Answer: Yes! The figure shows an example.

$$\begin{tikzcd} a \arrow[loop left] \arrow[r, bend left] & b \arrow[l, bend left] \arrow[loop right] \end{tikzcd}$$

Partial Orders

Definition

Let $(S, \preccurlyeq_S)$ and $(T, \preccurlyeq_T)$ be two posets. A **homomorphism** from $(S, \preccurlyeq_S)$ to $(T, \preccurlyeq_T)$ is a function $h : S \to T$ such that, for all $x, y \in S$,

$$x \preccurlyeq_S y \quad \text{implies} \quad h\,x \preccurlyeq_T h\,y.$$

Definition

Let $(S, \preccurlyeq_S)$ and $(T, \preccurlyeq_T)$ be two posets. An **order isomorphism** from $(S, \preccurlyeq_S)$ to $(T, \preccurlyeq_T)$ is a one-one correspondence $h : S \to T$ such that, for all $x, y \in S$,

$$x \preccurlyeq_S y \quad \text{iff} \quad h\,x \preccurlyeq_T h\,y.$$

Partial Orders

Definition

Let $(S, \preccurlyeq_S)$ and $(T, \preccurlyeq_T)$ be two posets. The **product of posets** S and T is the poset $(S \times T, \preccurlyeq)$ where the **product order** $\preccurlyeq$ is defined by:

For all $x_1, x_2 \in S$ and $y_1, y_2 \in T$,

$$(x_1, y_1) \preccurlyeq (x_2, y_2) \qquad \text{iff} \qquad x_1 \preccurlyeq_S x_2 \text{ and } y_1 \preccurlyeq_T y_2.$$

Outline

Relational Structures

Definition

Let L be a signature of a relational structure consisting of function and relation symbols, and let A and B be two L-structures. A **homomorphism** from A to B is a mapping h from the domain of A to the domain of B such that[†]

(i) for each n-ary function symbol F in L,

$$h(F^A\, x_1 \; \ldots \; x_n) = F^B\, (h\, x_1) \; \ldots \; (h\, x_n),$$

(ii) for each n-ary relation symbol R in L,

$$R^A(x_1, \ldots, x_n) \text{ implies } R^B(h\, x_1, \ldots, h\, x_n).$$

[†] From https://en.wikipedia.org/wiki/Homomorphism.

Outline

Topological Spaces

Definition

A **topology** on a set X is a collection τ of subsets of X such that

(i) $\emptyset$ and X are belong to τ,

(ii) the union of (finite or infinite) members of τ belongs to τ,

(iii) the intersubsection of finite members of τ belongs to τ.

The pair (X, τ) is a **topological space**.

Topological Spaces

Example

Let τ be the set of all open intervals in $\mathbb{R}$. The pair $(\mathbb{R}, \tau)$ is a topological space.

Topological Spaces

Example

Let τ be the set of all open intervals in $\mathbb{R}$. The pair $(\mathbb{R}, \tau)$ is a topological space.

Example

Let $\mathcal{P}\,S$ be the power set of the set S. The pair $(S, \mathcal{P}\,S)$ is a topological space.

Topological Spaces

Example

The pair $(\emptyset, \{\emptyset\})$ is a topological space.

Topological Spaces

Example

The pair $(\emptyset, \{\emptyset\})$ is a topological space.

Example

The pair $(\{*\}, \{\emptyset, \{*\}\})$ is a topological space.

Topological Spaces

Definition

Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function $f : X \to Y$ is **continuous** iff for all $V \in \tau_Y$, $f^{-1}(V) \in \tau_X$.

Definition

Let (X, τ) be topological space. A **base** (or **basis**) for τ is a collection $B \subset \tau$ such that every open set is a union of elements of B.

Topological Spaces

Definition

Let (X, τ_X) and (Y, τ_Y) be topological spaces. The **product topological space** of X and Y is the topological space $(X \times Y, \tau_{X \times Y})$, where $\tau_{X \times Y}$ is the topology generated by the Cartesian product $U_X \times U_Y \subset X \times Y$ of open sets $U_x \subset X$ and $U_Y \subseteq Y$.

Outline

Remark

The following definition was adapted from (Mac Lane 1971).

Axiomatic Category Theory

Remark

The following definition was adapted from (Mac Lane 1971).

Definition

Axiomatic Category Theory is the following two-sorted first-order theory with equality:

- The sorts of the theory are $\mathrm{Obj}()$ (*objects*), denoted by $A, B, C, \ldots$, and $\mathrm{Ar}()$ (*arrows*), denoted by $f, g, h, \ldots$.

(continued on next slide)

Axiomatic Category Theory

Definition (continuation)

- The undefined terms (language) of the theory are the *function* symbols[†]

$$
\begin{aligned}
\mathrm{dom} &: \langle \mathrm{Ar}(), \mathrm{Obj}() \rangle && (\textit{domain}),\\
\mathrm{cod} &: \langle \mathrm{Ar}(), \mathrm{Obj}() \rangle && (\textit{codomain}),\\
\mathrm{id} &: \langle \mathrm{Obj}(), \mathrm{Ar}() \rangle && (\textit{identity arrow}),
\end{aligned}
$$

  and the *relation* symbol

$$
\mathrm{comp} : \langle \mathrm{Ar}(), \mathrm{Ar}(), \mathrm{Ar}() \rangle \qquad (\textit{arrow composition}).
$$

(continued on next slide)

[†] The notation $\langle s_1, s_2, \ldots, s_n \rangle$ denotes a sort in many-sorted logic. See, for example, (Enderton 2001).

Axiomatic Category Theory

Definition (continuation)

Notation. An arrow f with $\operatorname{dom} f = A$ and $\operatorname{cod} f = B$ is written $f : A \to B$.

Notation. The arrow $\mathrm{id}(A)$ is denoted id_A.

(continued on next slide)

Definition (continuation)

- Non-logical axioms
  (i) For all arrows f and g, if $f : A \to B$ and $g : B \to C$ then there exists an unique arrow $h : A \to C$, such as $\mathrm{comp}(f, g, h)$.

  Notation. If $\mathrm{comp}(f, g, h)$ then the arrow h is denoted $g \circ f$.

  (ii) For all arrows f, g and h, if $f : A \to B$, $g : B \to C$ and $h : C \to D$ then

  $$h \circ (g \circ f) = (h \circ g) \circ f.$$

  (iii) For all object A, $\mathrm{dom}(\mathrm{id}_A) = \mathrm{cod}(\mathrm{id}_A) = A$.

  (iv) For all arrow f, if $f : A \to B$ then

  $$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f.$$

Remark

In general, would be incorrect to define categories as *models* of the previous two-sorted theory because, because *set theory models* would not include *large* categories.

Outline

References

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